

4.4 Method of Undetermined Coefficients

Let's consider the nonhomogeneous DE

$$a y'' + b y' + c y = g(x). \quad [NH]$$

The general solution of $[NH]$ has the

form :

$$y = y_c + y_p$$

where : $\begin{cases} y_c \rightarrow \text{complementary solution} \\ \quad \quad \quad (\text{or Homogeneous solution}) \\ y_p \rightarrow \text{A particular solution of } [NH]. \end{cases}$

In section 4.3, we showed how to find y_c when a, b, c are constants.

In this section, we will show a method to find y_p if $g(x)$ has the form of a sum of $[\text{Polynomial}] \times [\text{Exponential}] \times [\text{sinusoid}]$.

$$a y'' + b y' + c y = g(x)$$

Some examples of $g(x)$ that this section covers are :

$$g(x) = 10, \quad g(x) = 5x^2 + 3x, \quad g(x) = x e^{2x}$$

$$g(x) = (3x^2 + 1) e^{-5x} + x \sin(3x)$$

The method to find y_p works as follows.

Based on what $g(x)$ is, we make a "guess" about the form of y_p , leaving all coefficients undetermined.

Plug the guess into the DE to find the values of the coefficients.

Our basic guess of the form of y_p is

"The form of $g(x)$ and all possible derivatives of $g(x)$."

ex: ① solve $y'' - 3y' + 2y = 2x^2 + 1$

Step ① Find y_c

$$\text{Roots: } m^2 - 3m + 2 = 0 \rightarrow (m-2)(m-1) = 0 \\ \rightarrow m = 2, 1$$

$$y_c = c_1 e^{2x} + c_2 e^x$$

Step ② Find y_p

Since $g(x) = 2x^2 + 1$ is a quadratic Polynomial,
we look for y_p as a quadratic Polynomial.

$$y_p = Ax^2 + Bx + C \quad \leftarrow \text{our guess}$$

$$\Rightarrow y'_p = 2Ax + B$$

$$\Rightarrow y''_p = 2A$$

Plug y_p , y'_p , and y''_p into the DE.

$$[2A] - 3[2Ax + B] + 2[Ax^2 + Bx + C] = 2x^2 + 1$$

$$\Rightarrow \underline{2A}x^2 + \underline{(-6A + 2B)}x + (2A - 3B + 2C) = \underline{2}x^2 + \underline{0}x + \underline{1}$$

By matching the coefficients on both sides, we get:

$$\begin{cases} 2A = 2 \longrightarrow A = 1 \\ -6A + 2B = 0 \longrightarrow B = 3 \\ 2A - 3B + 2C = 1 \longrightarrow C = 4 \end{cases}$$

Then: $y_p = x^2 + 3x + 4$ is a particular solution of the DE.

The general solution is $y = y_c + y_p$

$$\Rightarrow \boxed{y = c_1 e^{2x} + c_2 e^x + x^2 + 3x + 4}$$

ex: ② Solve the IVP

$$\begin{cases} y'' - 5y' + 6y = 3e^{4x} \leftarrow \text{DE} \\ y(0) = 2 \\ y'(0) = 3 \end{cases} \leftarrow \text{initial conditions}$$

Step ① Find y_c

$$\begin{aligned} \text{Roots: } m^2 - 5m + 6 &= 0 \rightarrow (m-3)(m-2) = 0 \\ &\rightarrow m = 3, 2 \end{aligned}$$

$$y_c = c_1 e^{3x} + c_2 e^{2x}$$

Step ② Find y_p

Since $g(x) = 3e^{4x}$, we look for y_p as

$$y_p = A e^{4x}.$$

$$\Rightarrow y_p' = 4A e^{4x}$$

$$\Rightarrow y_p'' = 16A e^{4x}$$

Plug y_p , y'_p , and y''_p into the DE.

$$\left[16Ae^{4x} \right] - 5 \left[4Ae^{4x} \right] + 6 \left[Ae^{4x} \right] = 3e^{4x}$$

$$\Rightarrow 2Ae^{4x} = 3e^{4x}$$

$$\Rightarrow 2A = 3$$

$$\Rightarrow A = 3/2$$

$$\text{Then } y_p = \frac{3}{2} e^{4x}.$$

The general solution of the DE

$$\text{is } y = y_c + y_p.$$

$$y = c_1 e^{3x} + c_2 e^{2x} + \frac{3}{2} e^{4x}$$

Step ③ Find c_1, c_2

$$y = c_1 e^{3x} + c_2 e^{2x} + \frac{3}{2} e^{4x}$$

$$\Rightarrow y' = 3c_1 e^{3x} + 2c_2 e^{2x} + 6e^{4x}$$

$$y(0) = 2 \rightarrow c_1 + c_2 + \frac{3}{2} = 2$$

$$y'(0) = 3 \rightarrow 3c_1 + 2c_2 + 6 = 3$$

$$\Rightarrow \begin{cases} c_1 + c_2 = \frac{1}{2} \\ 3c_1 + 2c_2 = -3 \end{cases} \xrightarrow{\text{solve}} \begin{cases} c_1 = -4 \\ c_2 = \frac{9}{2} \end{cases}$$

The IVP solution is then:

$$y = -4e^{3x} + \frac{9}{2}e^{2x} + \frac{3}{2}e^{4x}$$

ex: ③ Solve $y'' - 6y' + 9y = 3\cos(2x)$.

Step ① Find y_c

$$\begin{aligned}\text{Roots: } m^2 - 6m + 9 &= 0 \rightarrow (m-3)^2 = 0 \\ &\rightarrow m = 3 \\ &\text{(single root)}\end{aligned}$$

$$y_c = c_1 e^{3x} + c_2 x e^{3x}$$

Step ② Find y_p

We might be tempted to guess the form of y_p as $A\cos(2x)$. Since all derivatives of $g(x) = 3\cos(2x)$ produce $\sin(2x)$ and $\cos(2x)$, we look for y_p as:

$$y_p = A\cos(2x) + B\sin(2x)$$

$$\Rightarrow y'_p = -2A\sin(2x) + 2B\cos(2x)$$

$$\Rightarrow y''_p = -4A\cos(2x) - 4B\sin(2x).$$

Plug y_p , y'_p , and y''_p into the DE and collect terms to get:

$$\begin{aligned} & (-4A - 12B + 9A) \cos(2x) + (-4B + 12A + 9B) \sin(2x) \\ & = 3 \cos(2x) + 0 \sin(2x) \end{aligned}$$

Match coefficients on both sides to get:

$$\begin{cases} -4A - 12B + 9A = 3 \\ -4B + 12A + 9B = 0 \end{cases} \Rightarrow \begin{cases} 5A - 12B = 3 \\ 12A + 5B = 0 \end{cases}$$

Solve the system: $\begin{cases} A = 15/169 \\ B = -36/169 \end{cases}$

The general solution is then: $y = y_c + y_p$.

$$y = c_1 e^{3x} + c_2 x e^{3x} + \frac{15}{169} \cos(2x) - \frac{36}{169} \sin(2x)$$

$$ay'' + by' + cy = g(x)$$

Here are some examples of $g(x)$ along with the basic guess of the form of y_p .

$g(x)$	form of y_p
5	A
$3x^2$	$Ax^2 + Bx + C$
$5x^3 + 2x - 3$	$Ax^3 + Bx^2 + Cx + D$
$7e^{-2x}$	Ae^{-2x}
$(5x + 3)e^{4x}$	$(Ax + B)e^{4x}$
$5\cos(2x)$	$A\cos(2x) + B\sin(2x)$
$e^x \sin(3x)$	$Ae^x \cos(3x) + Be^x \sin(3x)$
$x \cos(5x)$	$(Ax + B)\cos(5x) + (Cx + D)\sin(5x)$

Bad Cases

ex: ④ Solve $y'' + 4y = 3\cos(2x)$.

Step ① Find y_c

Roots : $m^2 + 4 = 0 \rightarrow m = \pm 2i$

$$y_c = c_1 \cos(2x) + c_2 \sin(2x)$$

Step ② Find y_p

since $g(x) = 3\cos(2x)$, like

example ③, we might be tempted

to look for : $y_p = A\cos(2x) + B\sin(2x)$

but this will not work since

y_p duplicates the form of y_c .

This y_p would give:

$$y_p'' + 4y_p = 0 \neq 3\cos(2x)$$

I refer to a situation like this as a Bad case.

The correct form of y_p is:

$$y_p = Ax \cos(ax) + Bx \sin(ax).$$

We multiplied the "basic guess" by x .

This form of y_p no longer duplicates any term in y_c .

$$y_p' = \underbrace{A \cos(ax) - 2Ax \sin(ax)}_{\text{product rule}} \dots$$

$$\dots + B \sin(ax) + 2Bx \cos(ax)$$

$$y_p'' = -4A \sin(ax) - 4Ax \cos(ax) + 4B \cos(ax) \\ \dots - 4Bx \sin(ax)$$

Plug these in the DE

$$y''_p + 4y_p = 3\cos(2x)$$

$$\begin{aligned} & \left[-4A\sin(2x) - 4Ax\cos(2x) + 4B\cos(2x) - 4Bx\sin(2x) \right] \\ & \dots + 4 \left[Ax\cos(2x) + Bx\sin(2x) \right] = 3\cos(2x) \end{aligned}$$

$$\begin{aligned} \Rightarrow & -4A\sin(2x) + 4B\cos(2x) \\ & \dots = 0\sin(2x) + 3\cos(2x) \end{aligned}$$

$$\Rightarrow \begin{cases} -4A = 0 \rightarrow A = 0 \\ 4B = 3 \rightarrow B = 3/4 \end{cases}$$

$$\Rightarrow y_p = \frac{3}{4} x \sin(2x)$$

The general solution is then:

$$y = y_c + y_p$$

$$y = c_1 \cos(ax) + c_2 \sin(ax) + \frac{3}{4} x \sin(ax)$$

$$a y'' + b y' + c y = g(x)$$

if $g(x) = g_1(x) + g_2(x) + \dots + g_k(x)$,

we apply the procedure separately to each $g_i(x)$ to find the Basic guess of the form of y_p .

$$y_p = y_{p_1} + y_{p_2} + \dots + y_{p_k}$$

For example if :

$$ay'' + by' + cy = \underbrace{x^2 + 4xe^{3x} + 5\cos(2x)}_{g(x) = g_1(x) + g_2(x) + g_3(x)}$$

Then, the basic guess for the form of y_p is :

$$y_p = \underbrace{Ax^2 + Bx + C}_{y_{p1}} + \underbrace{(Dx + E)e^{3x}}_{y_{p2}} + \underbrace{F\cos(2x) + G\sin(2x)}_{y_{p3}}$$

Bad Cases

If any of the y_{p_i} contains terms that duplicate terms in y_c , then y_{p_i} must be multiplied by x^n , where n is the smallest positive integer that eliminates the duplication.

ex: ⑤ Find the form of y_p in the following DEs.

$$(a) \quad y'' - 6y' + 8y = 5e^{2x} + x^2$$

$$(b) \quad y'' - 6y' + 9y = e^{2x} + 6e^{3x}$$

$$(c) \quad y'' + y' = x + 2 + 3xe^{-x}$$

$$(d) \quad y'' + 6y' + 13y = e^{-3x} \cos(2x) + \sin(2x)$$

Even though we want the form of y_p , we still have to find y_c to detect if we have a bad case.

(a) Roots: $m^2 - 6m + 8 = 0$
 $(m-2)(m-4) = 0$
 $m = 2, 4$

$$y_c = c_1 e^{2x} + c_2 e^{4x}$$

$$g(x) = 5e^{2x} + x^2$$

Basic guess of y_p

$$y_p = \underbrace{A e^{2x}}_{y_{p1}} + \underbrace{Bx^2 + Cx + D}_{y_{p2}}$$

Do we have a bad case? Yes, Ae^{2x} duplicates $c_1 e^{2x}$ from y_c .

$$\Rightarrow \boxed{y_p = Ax e^{2x} + Bx^2 + Cx + D}$$

↑
the correct form of y_p .

(b) Roots: $m^2 - 6m + 9 = 0 \rightarrow (m-3)^2 = 0$
 $\rightarrow m = 3$

$$y_c = c_1 e^{3x} + c_2 x e^{3x}$$

$$g(x) = e^{2x} + 6e^{3x}$$

Basic guess: $y_p = A e^{2x} + B e^{3x}$

Do we have a bad case? Yes.

Both $B e^{3x}$ and $B x e^{3x}$ duplicate

terms in y_c .

Then:

$$y_p = A e^{2x} + B x^2 e^{3x}$$

(c) Roots : $m^2 + m = 0$
 $m(m+1) = 0$
 $m = 0, -1$

$$y_c = c_1 + c_2 e^{-x}$$

$$g(x) = \underbrace{x+2}_{g_1(x)} + \underbrace{3xe^{-x}}_{g_2(x)}$$

Basic Guess

$$y_p = \underbrace{(Ax+B)}_{y_{p1}} + \underbrace{(Cx+D)e^{-x}}_{y_{p2}}$$

Do we have a bad case? Yes.

$\begin{cases} B \text{ duplicates } c_1 \\ D e^{-x} \text{ duplicates } c_2 \end{cases}$

$$\Rightarrow y_p = (Ax^2 + Bx) + (Cx^2 + Dx)e^{-x}$$

(d) Roots : $m^2 + 6m + 13 = 0$

$$m^2 + 6m + \textcircled{9} = -13 + \textcircled{9}$$

$$(m+3)^2 = -4$$

$$m+3 = \pm 2i \rightarrow m = -3 \pm 2i$$

$$y_c = e^{-3x} (c_1 \cos(2x) + c_2 \sin(2x))$$

$$g(x) = e^{-3x} \cos(2x) + \sin(2x)$$

Basic Guess

$$y_p = \underbrace{e^{-3x} (A \cos(2x) + B \sin(2x))}_{y_{p1}} + \underbrace{C \cos(2x) + D \sin(2x)}_{y_{p2}}$$

We have a Bad Case, y_{p1} duplicates y_c .

$$y_p = x e^{-3x} (A \cos(2x) + B \sin(2x)) + C \cos(2x) + D \sin(2x)$$

ex: ⑥ Solve the following IVP.

$$\begin{cases} y'' - y' - 6y = 3x + 2 + 5e^{-2x} \\ y(0) = 2 \\ y'(0) = 3 \end{cases}$$

Step ① Find y_c

Roots: $m^2 - m - 6 = 0$

$$(m - 3)(m + 2) = 0$$

$$m = 3, -2$$

$$y_c = c_1 e^{3x} + c_2 e^{-2x}$$

Step ② Find y_p

$$g(x) = \underbrace{3x + 2} + \underbrace{5e^{-2x}}$$

Basic Guess

$$y_p = Ax + B + Ce^{-2x}$$

Do we have a bad case? Yes.

Ce^{-2x} duplicates C_2e^{-2x}

$$\Rightarrow \boxed{y_p = Ax + B + Cxe^{-2x}}$$

Let's find A, B, C .

$$\begin{cases} y'_p = A + Ce^{-2x} - 2Cxe^{-2x} \\ y''_p = -4Ce^{-2x} + 4Cxe^{-2x} \end{cases}$$

Plug y_p, y'_p, y''_p in the DE to get:

$$\begin{aligned} & \left[-4c e^{-2x} + \cancel{4cx e^{-2x}} \right] - \left[A + c e^{-2x} - \cancel{2cx e^{-2x}} \right] \\ & \dots - 6 \left[Ax + B + \cancel{cx e^{-2x}} \right] = 3x + 2 + 5e^{-2x} \end{aligned}$$

Collect terms to get:

$$-6Ax + (-A - 6B) - 5c e^{-2x} = 3x + 2 + 5e^{-2x}$$

Match coefficients to get:

$$\begin{cases} -6A = 3 \longrightarrow A = -1/2 \\ -A - 6B = 2 \longrightarrow B = -1/4 \\ -5C = 5 \longrightarrow C = -1 \end{cases}$$

$$\therefore y_p = -\frac{1}{2}x - \frac{1}{4} - x e^{-2x}$$

The general solution is then:

$$y = y_c + y_p$$

$$y = c_1 e^{3x} + c_2 e^{-2x} - \frac{1}{2}x - \frac{1}{4} - x e^{-2x}$$

Step ③ Find c_1 and c_2

$$y' = 3c_1 e^{3x} - 2c_2 e^{-2x} - \frac{1}{2} - e^{-2x} + 2x e^{-2x}$$

$$y(0) = 2 \rightarrow c_1 + c_2 - \frac{1}{4} = 2$$

$$y'(0) = 3 \rightarrow 3c_1 - 2c_2 - \frac{1}{2} - 1 = 3$$

$$\Rightarrow \begin{cases} c_1 + c_2 = 9/4 \\ 3c_1 - 2c_2 = 9/2 \end{cases}$$

Solve to get : $c_1 = 9/5, c_2 = 9/20$.

The solution of the IVP
is then :

$$y = \frac{9}{5} e^{3x} + \frac{9}{20} e^{-2x} - \frac{x}{2} - \frac{1}{4} - x e^{-2x}$$